

## Case Study: Boosting Motor Performance with Smart Design

**Client:** Industrial Manufacturers & Automotive Engineering Firms

**Prepared by:** FEAAM GmbH

**Subject:** Increasing Torque and Efficiency in Axial Flux Machines (AFM)

### Executive Summary

Modern applications, from electric vehicles to industrial drones, require motors that are both powerful and compact. FEAAM GmbH has demonstrated a breakthrough in Axial Flux Machine (AFM) design by incorporating "flux barriers"—small, strategically placed gaps in the motor's stationary part, or stator. This design achieves 14% more torque and higher efficiency without increasing motor size or the cost of expensive raw materials like copper and magnets. FEAAM helps by optimizing internal magnetic paths to eliminate the traditional trade-off between high performance and production costs, allowing manufacturers to deliver superior power density while reducing energy waste.

### The Challenge: Getting More from Less

Traditional flat motors (Axial Flux Machines) are prized for being short and compact. However, they often suffer from "magnetic noise" (harmonics) that can cause energy loss and vibration.

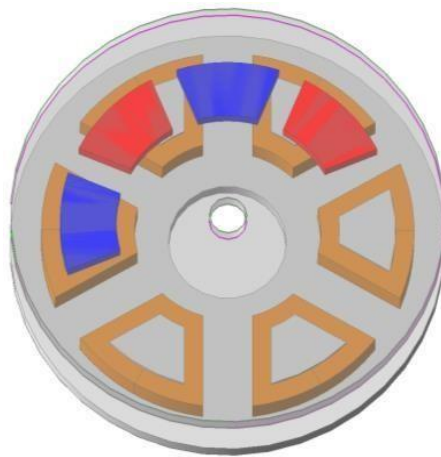


Fig. 1. Reference machine

To improve performance, manufacturers usually have two expensive options:

1. Increase the motor size, which adds weight.
2. Use more high-grade copper and magnets, which spikes production costs.

**The FEAAM Solution: The Flux Barrier Innovation**

Instead of adding more material, FEAAM engineers redesigned the internal magnetic paths. By adding flux barriers to the stator, we "steer" the magnetic field more effectively.

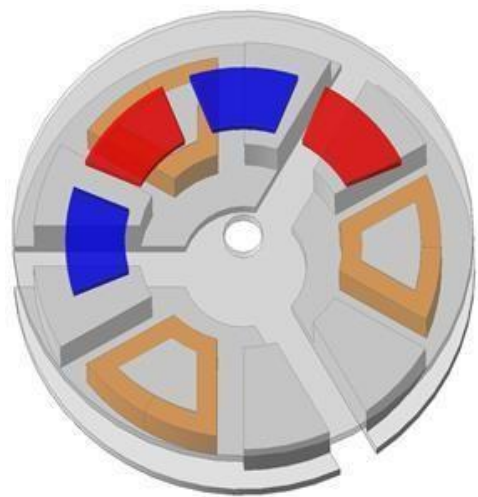


Fig. 2. Flux barrier machine

**How it works:**

Think of the magnetic field like traffic. In a standard motor, some "traffic" goes the wrong way, creating congestion (losses) and slowing things down. Our flux barriers act like highway dividers, forcing the magnetic "traffic" into the most productive lanes to generate power more efficiently.

**Measurable Results:**

We compared our new Flux Barrier (FB-AFM) design against a standard reference motor using high-precision Finite Element Analysis (FEA) simulations.

| Feature        | Standard Motor | FEAAM Improved Motor | Benefit        |
|----------------|----------------|----------------------|----------------|
| Torque (Power) | 15.96 mNm      | ~18.2 mNm            | 13.8% Increase |

| Feature        | Standard Motor | FEAAM Improved Motor | Benefit                |
|----------------|----------------|----------------------|------------------------|
| Efficiency     | 65.0%          | 67.7%                | Less Heat / More Range |
| Copper/Magnets | Baseline       | Same Amount          | No Extra Material Cost |
| Magnet Losses  | Baseline       | 25% Lower            | Longer Component Life  |

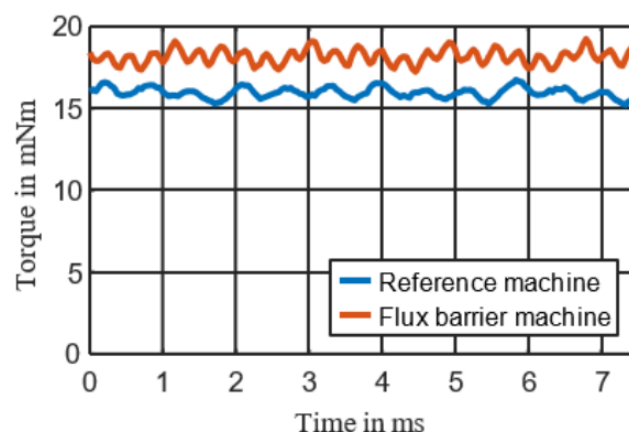


Fig.3. Comparison of the torque

### The Bottom Line for Your Business

This technology provides a massive competitive advantage for mass-manufactured goods. By choosing FEAAM's design approach, your company can:

- **Lower Production Costs:** Achieve your current torque requirements while using less copper and fewer magnets.
- **Go Greener:** Higher efficiency means less energy waste, fulfilling both ecological and economic goals.
- **Superior Power Density:** Get more power out of the same compact footprint.

FEAAM can assist your team by incorporating these strategically placed flux barriers into your axial motor designs, unlocking nearly 14% higher torque and improved efficiency without increasing the volume or cost of expensive active materials.